Digital Signal and Image Management

Exam project 2025-2026

Final grade

Exams consists of 2 parts:

- Assignments (30% of the final grade): attention to the deadlines!
  → To be conducted and delivered individually

- Project (70% of the final grade);
  → You are expected to work in groups of 2/3 people.
  → One delivery for each group
  Project consists of:
  - SW development
  - pseudo-demo
  - presentation (15 min.) to show the project and explain the implementation choices

Guidelines

The exam project consists of 3 parts, each covering one of the arguments macro-block covered in the lectures:

- Mono-dimensional signal processing
- Bi-dimensional signal processing
- Retrieval / Object detection /GAN

Guidelines

- Mono-dimensional signal processing:
  - The signal can be audio, physiologic, inertial, multi-channel, temporal series, etc.
- Bi-dimensional signal processing:
  - The signal can be both images or videos
- Retrieval/Object detection/GAN:
  - It is not required that it must be done on the same data of the previous tasks

Notes

- No basic outline is provided this year, so each group can propose their own theme, as long as the workload is commensurate with the course cfu
- It is therefore required to agree on the theme in advance with the lecturers
- The 3 tasks to be addressed may also be unrelated to each other

Notes

- No interface development is required
- The codes for the 3 modules can be just 3 different notebooks
- Input to the various modules (e.g., audio recording and photos) can be given manually, without having to implement automatic capture and transfer techniques
- The pseudo-demo must be shown at the exam and can be:
  - A real live demo (if the system allows it)
  - Video of a demo performed at home or in another environment (if the system does not allow it, e.g., too long computation time, etc.)